

HAMMAD AHMED

FPGA / RTL ENGINEER | EMBEDDED SYSTEMS & Digital Hardware

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Professional Profile

Electrical Engineering student and Embedded Systems Engineer with 5+ years of professional experience in embedded systems, digital electronics, industrial automation, PCB design, firmware, and IoT product development. Since 2026, developing practical experience in FPGA and RTL design through hands-on digital hardware projects, FPGA implementation, RISC-V processor design, and chip-design training. Experienced in designing synthesized Verilog RTL and implementing digital systems on physical FPGA platforms, including UART, VGA, ALU, pipelined arithmetic, APB, memory, FSMs, and counters. Currently developing expertise in SystemVerilog and ASIC design, with a strong focus on building a career in FPGA and RTL design while leveraging extensive embedded hardware experience.

FPGA, RTL & Embedded Skills

RTL & Digital Design

Verilog, RTL Design, Datapath & Control, FSMs, ALU, Pipelining, Registers, Counters, Memory Interfaces

FPGA

Arty A100T, Zybo Z7, F4PGA, PLL / Clock Generation, UART, VGA, SRAM, FPGA Peripherals

Processor Architecture

RISC-V RV32I, Single-Cycle Processor, Computer Architecture

Communication Protocols

APB, AHB, AXI, UART, I²C, SPI

Programming

Verilog, C, C++, Python, Scala, Chisel, RISC V Assembly

Embedded

STM32, ESP32, ESP8266, AVR, PIC, Firmware Development, Embedded Control Systems

Selected FPGA & RTL Projects

RISC-V RV32I Single-Cycle Processor

Chisel 3 / Scala 🐙 [GitHub](#)

- Designed a modular, synthesizable **RV32I-subset single-cycle, non-pipelined RISC-V processor** using Chisel 3 and Scala.
- Implemented the PC, instruction memory, register file, ALU, control unit, ALU control, immediate generator, and data memory.
- Implemented R-type instructions ADD, SUB, AND, OR, XOR, SLT; I-type ADDI and LW; S-type SW; B-type BEQ; U-type LUI; and J-type JAL.
- Compiled and executed RISC-V programs including Fibonacci, even/odd detection, register swapping, and multiplication-table operations.

UART Transmitter & Receiver

Verilog / FPGA

- Designed synthesizable UART TX/RX RTL and implemented the design on physical **Arty A100T** and **Zybo Z7** FPGA boards.
- Tested 9600-baud physical serial communication using a **100 MHz** Arty A100T system clock and **125 MHz** Zybo Z7 system clock.

VGA Display Controller

Verilog / Arty A100T

- Designed an FPGA-based VGA controller for display timing and RGB output generation.
- Used FPGA PLL resources to generate a **25 MHz pixel clock** and successfully displayed RGB patterns on an external monitor.

AMBA APB Interface with Custom Memory Slave

Verilog / FPGA

- Designed and implemented a Verilog-based APB interface supporting addressable read/write transactions.
- Developed a custom APB slave with integrated memory and physically validated the design on FPGA by writing data to selected slave addresses and reading the stored data back.

32-bit Pipelined Multiplier

Verilog / FPGA

- Designed a **32-bit, two-stage pipelined multiplier** using sequential datapath elements and pipeline registers.
- Implemented and tested the design on physical FPGA hardware.

32-bit ALU & Digital Controllers

Verilog / FPGA

- Designed and implemented a 32-bit ALU, up/down counter, single-port SRAM, FSM-based controllers, registers, and traffic-light controller.
- Physically implemented and tested the designs on FPGA hardware.

Professional Experience

Altronics Solutions

Full Stack Embedded Systems Engineer

Aug 2020 – Dec 2025

Karachi, Pakistan

- Progressed from Junior to Senior Engineer while independently managing complete embedded, industrial automation, IoT, and electronic product development projects.
- Designed embedded control systems, digital and analog circuits, schematics, PCBs, firmware, communication interfaces, and control architectures from concept through deployment.
- Developed a complete embedded control system for a fuel dispenser using ATmega32, including firmware, LCD interface, printer integration, configurable parameters, and localized system requirements.
- Designed segmented LCD display drivers for fuel-dispenser applications using different display topologies.
- Developed configurable IoT thermostat and humidistat controllers with local configuration and Android-based remote monitoring and control.
- Developed industrial monitoring and control systems using microcontrollers, RS-485, Modbus RTU, IoT connectivity, PLCs, and HMIs.
- Managed projects end-to-end, including requirements analysis, circuit design, PCB development, firmware, testing, troubleshooting, deployment, and technical coordination.

Education

Ziauddin University

B.E. Electrical Engineering / CGPA: 3.8/4.0

2022 – Dec 2026

Karachi, Pakistan

Final Year Project – Digital Controller for Electronic Jacquard Looms

In Progress

- Developing a digital controller that converts BMP-based textile designs into high-speed, synchronized solenoid control signals for an electronic Jacquard loom.
- Designing hardware and firmware for managing large-scale parallel I/O and synchronizing solenoid actuation with high-speed mechanical weaving cycles.
- Implementing digital interfaces, sensor feedback, communication, and power-failure recovery to improve system reliability.
- Expected completion: **Nov 2026**; B.E. Electrical Engineering expected graduation: **Dec 2026**.

Jinnah Polytechnic Institute

Diploma of Associate Engineer (DAE), Electronics / A-1, 83.8%

2019 – 2022

Karachi, Pakistan

Chip Design & Hardware Training

MERL Chip Design – Professional Development Course

Completed Feb 2026

- Completed a two-month chip-design program covering Scala, Chisel 3, RISC-V architecture, and hardware construction.
- Designed a complete RV32I-subset single-cycle processor and developed its datapath and control architecture using Chisel.
- Executed compiled RISC-V programs and analyzed processor behavior using simulation and waveform analysis.

PSEB Chip Design Program

In Progress

- Coursework includes Digital Logic Design, Verilog, SystemVerilog, communication protocols, FPGA design, and ASIC design.
- Currently progressing through SystemVerilog and ASIC design topics.

Selected Industrial & Embedded Projects

- **GENSET Monitoring & Control:** Developed IoT-based monitoring, scheduling, and control systems for 300 kVA, 250 kVA, 100 kVA, and 50 kVA generator systems across multiple sites.
- **3-Phase Energy Monitoring:** Developed IoT-enabled three-phase energy monitoring and remote data acquisition systems.
- **RO Plant Monitoring:** Developed an IoT-based monitoring solution for industrial reverse-osmosis plant parameters.
- **Damper Control:** Developed a temperature-based damper control system with HMI interface.
- **LUX & Air Quality Meter:** Developed an RS-485 Modbus sensing system with a centralized IoT hub.
- **AC Control & Scheduling:** Developed IoT-based AC controllers and centralized RS-485 Modbus hubs for multi-location control and scheduling.
- **Fuel Dispenser Systems:** Developed embedded fuel-dispenser control systems, segmented display drivers, and IoT-based fuel monitoring solutions.

Selected Product & Hardware Development

- **Electronic Fabric Take-Up Controller:** Engineered the complete circuit and PCB, developed an RS-485 Modbus HMI, and implemented firmware for fabric pick-density and warp-tension synchronization in industrial looms.
- **Electronic Weft Color Control System:** Designed the complete circuit and industrial PCB and developed low-latency firmware for synchronizing yarn-color sequences with high-speed weaving cycles.
- **Digital IC Tester:** Designed a Python-based GUI and hardware system for testing real-world digital integrated circuits.
- **Optical Wireless Transceiver:** Designed a transceiver for transmitting and receiving images through optical wireless communication.
- **Force Myography Prosthesis:** Worked on development of an FMG-controlled upper-limb prosthetic arm.

Leadership, IEEE & Technical Outreach

IEEE Ziauddin University Student Branch

May 2026 – Present

Chairperson

Karachi, Pakistan

- Lead the student branch executive committee in planning, coordinating, and executing technical activities and programs.
- Oversaw the organization of **16 technical webinars** during the 2026 summer program, coordinating teams, speakers, and event operations.

IEEE Ziauddin University Student Branch

Aug 2024 – Dec 2025

General Secretary

Karachi, Pakistan

- Managed documentation, records, event operations, team coordination, and organizational activities.
- Contributed to the planning and execution of **38 IEEE Student Branch events** across 2024–2025, including technical workshops, competitions, and student activities.
- Supported planning and execution of technical events, workshops, and student development activities.

Technical Workshops & Outreach

- **Getting Started with Arduino and Wokwi** – Conducted a hands-on workshop for first-year students at Ziauddin University, Nov 2024.
- **How IoT Has Changed the World?** – Conducted a two-day technical workshop for students in the Biomedical Engineering Department at Ziauddin University, Sep 2023.
- **IEEEExtreme 19.0 Ambassador** – Volunteered as an IEEE Student Branch Ambassador for the IEEEExtreme 19.0 programming competition, Oct 2025.

Recognition

- **Token of Appreciation** for conducting “Getting Started with Arduino and Wokwi”, Nov 2024.
- **Memento of Appreciation** for conducting the two-day “How IoT Has Changed the World?” workshop, Sep 2023.

Certifications & Achievements

- **MERL Chip Design Professional Development Course**, completed Feb 2026.
- **1st Position in Department and 2nd Position in Institute**, DAE Electronics, 2022.
- **2nd Position**, Xtreme Encode, ENCIDE MACE in collaboration with IEEEExtreme 18.0 Region 10, 2024.
- **2nd Position**, Inter-Departmental Project Competition, Ziauddin University, 2023.
- **Industrial Automation (Professional)**, AutoCon, 2022.
- **Industrial Automation Level-2**, NAVTTC, 2021.

Languages

English | Urdu